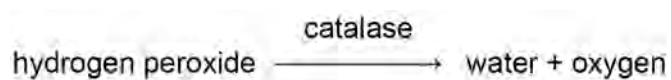


0 6

The reaction catalysed by the enzyme catalase is shown below.

**0 6 . 1**

Use your knowledge of the induced-fit model to explain how catalase breaks down hydrogen peroxide.

[3 marks]



Potato cells contain catalase.

A student investigates the effect of the surface area of potato pieces on the rate of the reaction of catalase.

0 6 . 2

Before starting the investigation, the student reads the following information.

Hydrogen peroxide:

- is a clear, colourless solution
- is harmful if swallowed
- causes serious eye damage
- may cause damage or irritation to the skin
- may cause respiratory irritation.

Use this information to suggest **two** safety precautions the student should take.

Give **one** reason for each safety precaution.

[2 marks]

Precaution 1 _____

Reason _____

Precaution 2 _____

Reason _____

Question 6 continues on the next page

Turn over ►

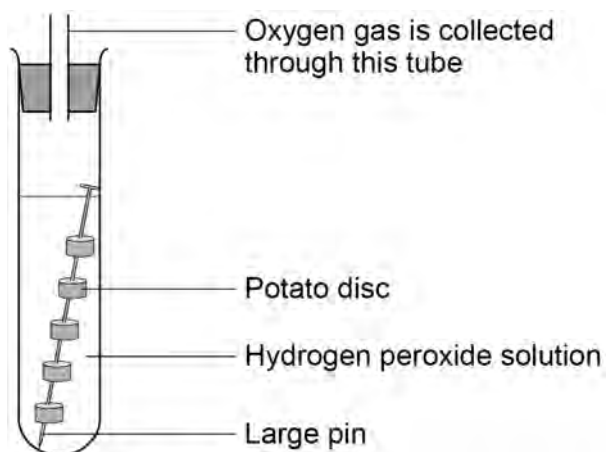


The student uses the following method.

1. Cut a 75 mm long cylinder of potato into 5 discs of equal size.
2. Arrange the discs on a large pin in a boiling tube.
3. Add 40 cm³ of 1.0 mol dm⁻³ hydrogen peroxide.
4. Record the volume of oxygen collected in 2 minutes.
5. Repeat steps 1 to 4 with an uncut cylinder of potato 75 mm long.

Figure 10 shows the equipment the student uses.

Figure 10



0 6 . 3

Suggest why the student arranges the potato discs on the pin as shown in **Figure 10**.
[1 mark]



Table 3 gives measurements for one potato disc and the uncut potato cylinder used in the investigation.

Do not write
outside the
box

Table 3

	Length (L) / mm	Diameter / mm
One potato disc	15	10
Uncut potato cylinder	75	10

0 6 . 4

Calculate how many times greater the total surface area of the five discs of potato is compared with the surface area of the uncut cylinder.

Use the information in **Table 3** and the formula below.

The formula for the surface area of a disc or cylinder = $2\pi r(r+L)$

where

r = radius

L = length

$\pi = 3.14$

[3 marks]

_____ times greater

0 6 . 5

The rate of reaction is faster with the five discs of potato than with the uncut potato cylinder.

Explain why.

[1 mark]

10

Turn over ►

